

Yongxi Feng (冯泳熙)

The Chinese University of Hong Kong, Shenzhen Email: 123090113@link.cuhk.edu.cn Phone: +86 180-2236-0783

Personal Website LinkedIn GitHub Languages: Mandarin Chinese (native) | Cantonese (native) | English (fluent)

Note: Colored entries are hyperlinked to supporting documentation, code repositories, or official records..

Education

The Chinese University of Hong Kong, Shenzhen (CUHK-SZ), Shenzhen, China

B.S. in Statistics

August 2023 – June 2027

CGPA: 3.879/4.0 Major GPA: 3.926/4.0 Rank: 2/67 (major), 18/361 (School of Data Science)

- **Dean's List Award:** AY2023–24, AY2024–25, AY2025–26 (three consecutive years)
- **Academic Performance Scholarship:** Class C (AY2024–25); Class A (AY2025–26)
- Core Courses: Honors Probability & Statistics, Linear Models, Machine Learning, Reinforcement Learning, Time Series Analysis, Stochastic Processes, Optimization, Mathematical Analysis

University of British Columbia, Faculty of Science, Vancouver, BC, Canada

Exchange Student

January 2026 – April 2026

Research Experience

Doctor-Aligned Clinical Decision Modeling with Latent State Calibration, CUHK-Shenzhen

Undergraduate Researcher (Advisor: Prof. Shuang Li)

2025 – Present

- Investigated doctor-aligned clinical decision modeling from longitudinal EHR trajectories, focusing on how physicians may interpret the same patient evidence differently due to heterogeneous practice styles, preferences, and reasoning patterns
- Designed a dual-worldview architecture that separates an objective patient latent state from a doctor-specific latent reasoning profile learned from historical clinical behavior
- Proposed shared high-level latent calibration rules that transform the objective patient representation into a doctor-perceived state before action prediction, avoiding brittle hand-written predicates and doctor-specific rule banks

Wearable Sensor Anomaly Detection —LLM-Based Framework, CUHK-Shenzhen

Undergraduate Researcher (Advisor: Prof. Kaixin Zhou)

Summer 2025

- Investigated the clinical validity gap in wearable health data: existing anomaly detection approaches rely on population-level statistical thresholds that conflate device artifacts with subject-specific physiological deviations
- Processed and tokenized time-series physiological signals from NHANES (2,000+ participants) into structured sequences suitable as LLM inputs; designed prompt engineering and fine-tuning strategies to adapt LLMs for anomaly detection over irregular temporal data
- Implemented a teacher-student framework with knowledge distillation, using FRIM-based statistical soft labels to jointly learn population-level and individualized anomaly patterns

Independent Study: Deep Active Inference & Cognitive Modeling, CUHK-Shenzhen

Participant (Organized by Dean Jianhua Huang, School of AI; co-studied with PhD mentors)

2024 – 2025

- Selected for faculty-initiated reading and research group; studied foundational literature at the intersection of active inference, the Free Energy Principle, and deep learning architectures for cognitive modeling

Industry Experience

The Story Behind the Photo: A Multimodal Interview and Memoir Generation System, Tencent Collaboration

Research Engineer (Advisor: Prof. Zhongxiang Dai)

[\[GitHub\]](#) [\[Verification\]](#)

2025 – Present

- Built multimodal system enabling visually grounded oral history collection: users upload photographs; the system analyzes visual content (objects, architectural details, etc.) and generates contextually targeted interview questions
- Designed and implemented the **question generation pipeline**: extracting semantically meaningful visual features and mapping them to coherent interview prompts; developed cross-image context management to enable longitudinal narrative construction
- Developed simulation interview corpus for system evaluation; conducted **benchmark evaluation** across multiple LLMs (Gemini, Hunyuan, Tencent Yuanbao) on question relevance, narrative coherence, and story richness

- Received **Meritorious Presentation Award** (top 10 among 54 teams)

Context-Aware Music Scene Recommendation, *Beijing Qiyin Technology*

Machine Learning Engineer Intern (*Mentor: Zhongxiang Luo, CTO of Qiyin Technology*) *May 2025 – Present*

- Built context-aware music recommendation POC ranking 18 scenarios via missing-aware rules, semantic embeddings, and personalized preference learning
- Designed FastAPI backend with SQLite feedback history, supporting Top-K recommendations and online updates from listening behavior, skips, and user corrections
- Added hierarchical habit modeling and geospatial clustering to learn user routines while remaining robust to missing permissions and noisy mobile sensor data

Teaching & Academic Service

Undergraduate Teaching Assistant (USTF), MAT3007 Optimization, *CUHK-Shenzhen*

(Instructor: Prof. Xiao Li) **[Verification]** *Fall 2025*

- Conducted three tutorial sessions decomposing convex optimization concepts for mixed-background cohorts; held weekly office hours with individualized problem-solving support; evaluated assignments and provided written feedback

Peer Academic Advisor, School of Data Science, *CUHK-Shenzhen*

(Coordinator: Prof. Dan Li) **[Verification]** *AY2025–2026*

- Selected as outstanding peer advisor; conducted one-on-one advising for 20+ students on academic planning, research entry, and graduate application strategy; organized information sessions on major pathways

Panelist, Undergraduate Academic Exchange Session, *CUHK-Shenzhen SDS*

(Referee: Prof. Shuang Li) **[Event]** *Fall 2025*

- Selected as student speaker to share experiences in research entry and study abroad

Review Session Instructor, MAT3007 Optimization, *Minerva College, CUHK-SZ*

- Initiated and delivered exam preparation lectures in coordination with the residential college *Spring 2024*

Cross-Disciplinary Data Science Consultant, *Minerva College, CUHK-Shenzhen*

(Collaboration with Clinical Medicine Students) **[Outstanding Project Award]** *Spring 2024*

- Conducted hands-on training on statistical interpretation and public health recommendation formulation; co-developed prevention and resource allocation strategies; presented findings via academic poster
- Received **Outstanding Capstone Project Award** (Minerva College)

Volunteer English Teacher, Rural Primary School, *Guangxi, China*

- Designed and delivered English curriculum for students with heterogeneous proficiency levels *Spring 2023*

Selected Course Projects

Time Series Analysis (STAT443): *Forecasting Daily Influenza Incidence in Kawasaki City* — Led 4-person team; proposed research design; implemented ARIMAX models with lagged meteorological covariates; seasonal autoregressive structure dominates predictability, with meteorological variables contributing less than 1% RMSE reduction.

Machine Learning (DDA3020): *Hybrid Trading Framework for EMH Testing (Kaggle)* — Decoupled LightGBM directional prediction from PPO volatility-adaptive position sizing; implemented rolling-window temporal splits to prevent leakage; improved Kaggle score via systematic hyperparameter optimization.

Reinforcement Learning (DDA4230): *Credit Assignment in Closed Cooperative Systems* — Systematic survey of MARL credit assignment paradigms (VDN, QMIX, QTRAN, COMA, MADDPG, TarMAC); analyzed theoretical assumptions and practical trade-offs in cooperative settings.

Technical Skills

Programming & Tools

- Python (PyTorch, scikit-learn, pandas, NumPy)
- R (tidyverse, survival, ggplot2)
- SQL, MATLAB, Stata, $\LaTeX$
- Git / GitHub

Research Interests: Machine learning reliability and clinical validity in healthcare data; wearable/sensor time-series modeling; uncertainty quantification; bridging algorithmic anomaly detection and clinical interpretation.

Methods

- Statistical learning, time-series analysis
- Multiple imputation, missing data methods
- Reinforcement learning, LLM fine-tuning
- Uncertainty quantification (introductory)